

Fine-Tuning Open-Source Large Language Models for Industrial Quality Assurance

A Practical Guide for Small and Medium Enterprises

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Abstract

We present a practical methodology for fine-tuning open-source large language models (LLMs) on domain-specific quality-assurance tasks encountered by small and medium enterprises (SMEs) in the manufacturing sector. Using LoRA (Low-Rank Adaptation) applied to Qwen-2.5-7B-Instruct and Llama-3.1-8B-Instruct, we demonstrate that a fine-tuned 7-8 billion parameter model running on a single NVIDIA RTX 4090 (24 GB VRAM) can match or exceed the performance of a general-purpose 70B model on material-certificate compliance checking, achieving an F1 score of 0.94 versus 0.89 for the base 70B model. Training required 12,500 synthetic examples generated via GPT-4o and validated by rule-based checkers, with total compute cost under EUR 200 on a rented A100-80GB instance from Lambda Labs. We release the training pipeline, evaluation harness, and adapter weights under open licences to support reproducibility.

1. Introduction

The adoption of AI-powered automation in manufacturing SMEs remains constrained by three practical barriers: (1) data sovereignty requirements that prohibit sending proprietary documents to cloud APIs, (2) limited IT budgets that rule out enterprise-grade GPU clusters, and (3) a shortage of ML engineering talent to build and maintain custom pipelines. A 2025 survey by the European Association of Machine Tool Industries (CECIMO) found that only 18% of manufacturing SMEs with fewer than 250 employees had deployed any form of generative AI in production workflows, compared to 67% of firms with more than 1,000 employees.

Recent advances in parameter-efficient fine-tuning (PEFT) methods -- particularly LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023) -- have dramatically lowered the compute requirements for specialising large models. A Qwen-2.5-7B model can be LoRA-tuned on a single 24 GB GPU in under 4 hours for a dataset of 10,000-15,000 examples, producing adapter weights of approximately 50 MB that preserve the base model's general capabilities while adding domain-specific competence.

This report describes a complete fine-tuning pipeline developed during a consulting engagement with Meridian Aerospace Components Pvt. Ltd. (Pune, India; ~180 employees), an AS9100D-certified manufacturer of precision turbine components. The target task -- automated comparison of incoming material certificates against customer-imposed specifications -- represents a high-frequency quality-assurance workflow that consumes approximately 6 hours per week of a senior quality engineer's time at Meridian.

2. Methodology

2.1 Task Definition

The compliance-checking task requires comparing a supplier material certificate (typically a 1-2 page PDF listing chemical composition, mechanical properties, and heat-treatment records) against a customer specification document that defines acceptable ranges for each property. The model must flag every deviation -- where a certificate value falls outside the specification's permitted range -- with a structured justification.

We formalized this as a structured-output task where the model receives the full text of both documents and must return a JSON object listing each flagged property with fields: `property_name`, `spec_requirement`, `cert_value`, `deviation_type` (`out_of_range` | `missing` | `format_mismatch`), and justification. The ground truth was established by a senior quality engineer at Meridian, Mr. Anand Kulkarni (15 years experience, AS9100D Lead Auditor certified), who manually reviewed and labeled 200 real certificate-specification pairs.

2.2 Training Data Generation

Direct use of Meridian's proprietary certificates was prohibited by their data governance policy (ISO 27001 Annex A.8 compliance). Instead, we generated 12,500 synthetic certificate-specification pairs using GPT-4o (accessed via the OpenAI API, version `gpt-4o-2024-08-06`) with the following pipeline:

Step 1: Material template generation. We defined 35 material families (e.g. Inconel 718, Ti-6Al-4V, AISI 304L, AA7075-T6) with realistic property distributions sourced from ASM International's Materials Database and MMPDS-17 (Metallic Materials Properties Development and Standardization, revision 17, published January 2024 by Battelle Memorial Institute).

Step 2: Specification synthesis. For each material, GPT-4o generated 50 plausible customer specifications with varying strictness levels, modelling real patterns from aerospace primes such as Rolls-Royce plc (Derby, UK), Safran SA (Paris, France), and Pratt & Whitney (East Hartford, Connecticut, USA).

Step 3: Certificate synthesis with injected deviations. Each certificate was generated with 0-4 deliberate deviations drawn from a taxonomy of 12 deviation types. Deviations were validated by a rule-based checker (Python, ~800 lines) that verified the flagged values genuinely fell outside the specification ranges.

Step 4: Human validation. A random 10% sample (1,250 pairs) was reviewed by Anand Kulkarni and Ms. Priya Sharma (Quality Engineer, 8 years experience) at Meridian, achieving inter-rater agreement of Cohen's kappa = 0.91 on deviation labels.

2.3 Fine-Tuning Configuration

We fine-tuned two base models using the Unsloth library (version 2024.12, developed by Daniel and Michael Han, unsloth.ai):

Model A: Qwen-2.5-7B-Instruct (Alibaba DAMO Academy, released November 2024) Model B: Llama-3.1-8B-Instruct (Meta AI, released July 2024)

LoRA hyperparameters (identical for both models): - Rank (r): 32 - Alpha: 64 - Dropout: 0.05 - Target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj - Learning rate: $2e-4$ (cosine schedule with 100-step warmup) - Batch size: 4 (gradient accumulation steps: 8, effective batch size: 32) - Epochs: 3 - Max sequence length: 4,096 tokens - Quantization: 4-bit NormalFloat (NF4) via bitsandbytes

Training was conducted on a single NVIDIA A100-80GB GPU rented from Lambda Labs (San Francisco, CA) at a rate of USD 1.29/hour. Model A completed training in 3 hours 42 minutes (total cost: USD 4.78). Model B completed in 4 hours 11 minutes (total cost: USD 5.39). The resulting adapter weights were 48 MB (Model A) and 52 MB (Model B).

3. Results

Evaluation was conducted on a held-out test set of 500 certificate-specification pairs (none seen during training). Each pair was run 10 times at temperature 0.1 to assess output stability. The following table summarises performance:

Model	Prec.	Recall	F1	Std(F1)	Lat.(s)	GPU VRAM
Qwen-2.5-72B (baseline)	0.88	0.91	0.89	0.04	8.3	~40 GB
Llama-3.3-70B (baseline)	0.85	0.90	0.87	0.05	9.1	~40 GB
Qwen-2.5-7B + LoRA (ours)	0.93	0.95	0.94	0.02	1.4	~6 GB
Llama-3.1-8B + LoRA (ours)	0.90	0.93	0.91	0.03	1.6	~6 GB
Qwen-2.5-7B (no fine-tune)	0.62	0.58	0.60	0.11	1.3	~6 GB

Key findings: (1) The fine-tuned Qwen-2.5-7B model achieved F1 = 0.94, outperforming the general-purpose 72B model (F1 = 0.89) while requiring 85% less VRAM. (2) Output stability improved substantially -- Std(F1) dropped from 0.04 to 0.02, meaning the fine-tuned model produces more consistent outputs across repeated runs. (3) Inference latency decreased from 8.3s to 1.4s, a 5.9x speedup attributable to the smaller model size.

4. Discussion and Cost Analysis

The total project cost breaks down as follows:

- Synthetic data generation (OpenAI API): USD 187.40 (12,500 pairs x ~750 input + ~500 output tokens each, at GPT-4o pricing of USD 2.50/M input + USD 10.00/M output as of February 2026) - Human validation (Meridian QA team, 40 hours at internal rate): not billed externally - GPU rental for training (Lambda Labs A100-80GB, 8 hours): USD 10.32 - Evaluation runs (same GPU, 3 hours): USD 3.87 - Total external compute cost: USD 201.59

For deployment, Meridian purchased a single NVIDIA RTX 4090 workstation (assembled by a local system integrator in Pune, total cost INR 3,25,000 / approximately EUR 3,600) running Ubuntu 22.04 LTS with vLLM (version 0.6.3) as the inference server. The workstation serves the fine-tuned Qwen-2.5-7B model at an average throughput of 45 tokens/second, sufficient for the 6-8 certificate checks processed per day.

Projected annual savings: the quality engineer previously spent ~6 hours/week on manual certificate review. With the AI assistant handling first-pass screening (human reviews only flagged deviations), this reduced to ~1.5 hours/week. At a fully-loaded labour cost of approximately EUR 18/hour for a quality engineer in Pune, the annual saving is approximately EUR 4,200 -- yielding a payback period of under 12 months on the hardware investment.

5. Limitations

Several limitations should be noted: (1) The training data is entirely synthetic. While validated against real-world patterns by domain experts, subtle distributional differences between synthetic and real certificates may affect field performance. (2) The evaluation uses N=10 runs per test case; larger N would yield tighter confidence intervals. (3) Model versioning: both Qwen-2.5 and Llama-3.1 may receive silent updates from their providers. Our results are locked to the specific checkpoints downloaded on 10 January 2026 (Qwen: commit hash 3f8c2a9e; Llama: commit hash 7d1b4f5c from Hugging Face Hub). (4) The pipeline has been validated only on English-language certificates. Extension to German and French certificates -- common in European aerospace supply chains -- is planned for Q3 2026, in collaboration with RWTH Aachen University (Prof. Dr. Katharina Wendt, Chair of Quality Management and Manufacturing Metrology).

References

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